to withstand stress, certain congenital diseases like diabetes and hypertension.

4. DNA is the source of information for the synthesis of all cellular proteins. The segment of DNA that contains information for a protein is known as gene.

7.8 Ribonucleic acids (RNA)

RNAs, the second most abundant form of nucleic acids inside the cell are single-stranded and composed of nucleotides with ribose as their sugar. The major role of RNA is in protein synthesis as it aids in transfer of information from nucleus to cytosol, decoding, and synthesis of proteins. RNA nucleotides contain three components.

- Nitrogenous Base
- Ribose Sugar
- Phosphate Group

The nitrogenous bases include adenine (A), guanine (G), cytosine (C) and uracil (U).

Eventhough RNA is single-stranded, it has the ability to form three-dimensional structures like hairpin loops by base pairing within the strand. Adenine pairs with uracil (A=U) and guanine pairs with cytosine (G≡C). Chargaff's rule is not followed by RNA.

7.8.1 Types of RNA

There are three major types of RNAs in all prokaryotic and eukaryotic cells. They are (1). Messenger RNA (mRNA). (2). Transfer RNA (tRNA) (3). Ribosomal RNA (rRNA). They differ from each other by size, shape, formation and stability.

i. Messenger RNA

They are named so because they carry the information from nucleus to cytosol. It accounts for 1-5% of cellular RNA. They do not have specific secondary structure like helices. They are single stranded linear molecules. They consist of 1000-10,000 nucleotides. They have a free or phosphorylated 3′and 5′end. They have different life span ranging from few minutes to days.

Prokaryotic mRNA is different from eukaryotic mRNA. Prokaryotic mRNAs are polycistronic, that is they code for many proteins. While eukaryotic mRNAs are monocistronic and contain exons (expressed sequences) and introns (intervening sequences). Eukaryotic mRNAs are capped at 5′end by methylated guanosine triphosphate. Capping protects mRNA from nuclease attack. At 3′end, a polymer of adenylate (polyA) is found as tail. Poly A tail protects mRNA from nuclease attack.

Intrastrand base pairing among complementary bases allows folding of the linear molecule. As a result, haripin or loop like secondary structure is formed.

Functions

1. mRNA is a direct carrier of genetic information from the nucleus to cytoplasm.
2. It contains information required for the synthesis of protein molecules.

ii. Transfer RNA

It accounts for 10-15% of total cell RNA. Usually they consist of 50-100 nucleotides. They are single stranded

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